

Numerical Methods [CH-2-16 (MO)]

Mid Semester Examination

Maximum Marks: 20

Maximum Duration: 90 Minutes

1. Define round off error and truncation error with an example? (2 Marks)

2. Suppose that you have the task of measuring the lengths of a bridge and a rivet and come up with 9999 and 9 cm, respectively. If the true values are 10,000 and 10 cm, respectively, then estimate the percent relative error in each case. (2 Marks)

3. Which one of the following option is true? (1 Mark)

I. Regular falsi method, II. Tangent method, III. Halving method, IV. Secant method
(a) Bracket method (b) Open method

1. I(a), II(a), III(b), IV(b)
2. I(a), II(b), III(b), IV(a)
3. I(b), II(a), III(b), IV(a)
4. I(a), II(b), III(a), IV(b)

4. Solve and then identify the Newton-Raphson iteration scheme for finding the square root of 2. [Note: No marks for identifying/noting only the option, full proof is must] (2 Marks)

(a) $x_{n+1} = \frac{1}{2} \left(x_n - \frac{2}{x_n} \right)$ (b) $x_{n+1} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right)$ (c) $x_{n+1} = \frac{2}{x_n}$ (d) $x_{n+1} = \sqrt{2 + x_n}$

5. Use LU decomposition method to find the solution for the following system of equations. (4 Marks)

$x_1 + x_2 + x_3 = 1$; $4x_1 + 3x_2 - x_3 = 6$; $3x_1 + 5x_2 + 3x_3 = 4$

6. For the special case of symmetric, positive definite matrices, the Cholesky decomposition method is given by

$$l_{kk} = \sqrt{a_{kk} - \sum_{j=1}^{k-1} l_{kj}^2} \quad ; \text{ and } \quad l_{ki} = \frac{1}{l_{ii}} \left(a_{ki} - \sum_{j=1}^{i-1} l_{ij} l_{kj} \right) \quad \text{for } k > i$$

Using the above, compute the L matrix for the following matrix A.
What is the U matrix in this case? (3 Marks)

$$A = \begin{bmatrix} 4 & 12 & -16 \\ 12 & 37 & -43 \\ -16 & -43 & 98 \end{bmatrix}$$

7. Use Gauss-Seidel method for the following set of equations with an initial guess (0,0,0) and find the values of c_1 , c_2 , and c_3 after 1st and 2nd iterations, up to two decimal places.

$15c_1 - 3c_2 - c_3 = 3300$; $-3c_1 + 18c_2 - 6c_3 = 1200$; $-4c_1 - c_2 + 12c_3 = 2400$ (4 Marks)

8. Find the functional value and the Jacobian of the nonlinear system of equations $y = -x^2 + x + 0.75$ and $y + 5xy = x^2$ at the value $x=y=1.2$. (2 Marks)

$$J(x, y) = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix}_{(1.2, 1.2)}$$

Note: Marks will not be awarded if only final answer is given without the intermediate steps [Except for Q. (3)]